Cảm biến và đo lường cho robot (RBE3042)

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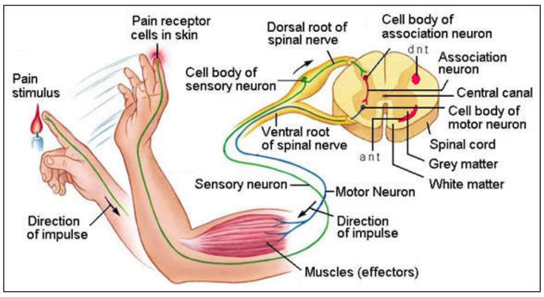
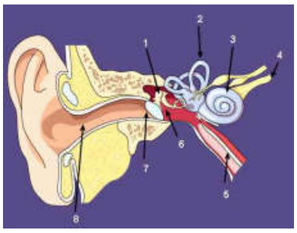
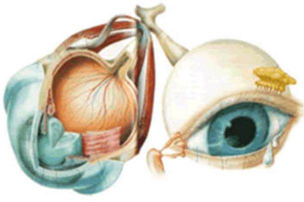
Human and robot sensors quiz

- How many sensors or senses do humans have? What are they?
- Describe how any two of those human sensor work
- Give at least three examples of robot sensors that are similar to human senses.

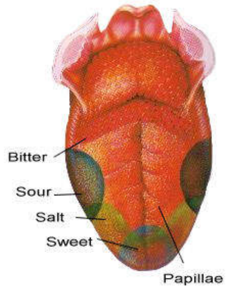




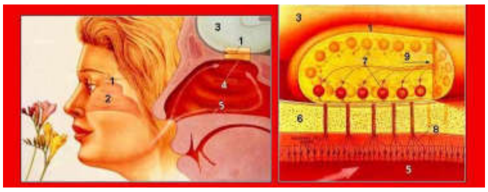
Auditory system 1.Ossicle. 2. Semicircular canal. 3. Cochlea. 4. Auditory nerve 5. Eustachian tube 6.Midle ear. 7. Ear drum 8. External ear canal



Light enters the front of the eye through the pupil and is focused by the lens onto the retina. Rod cells on the retina respond to the light and send a message through the optic nerve fiber to the brain.



The tongue is covered with dozens of pimple-like projections called papillae. These grip and move food when you chew. Around the sides of the papillae are about 10,000 microscopic taste buds. Different parts of the tongue are sensitive to different flavours: sweet, salt, sour and bitter.



Olfactory system. 1. Olfactory bulb. 2. nasal cavity. 3. Brain. 4. Olfactory epithelium 5. Vomeronasal organ. 6. Ions. 7 Glomeruli. *. Axon. 9. To olfactory cortex

The sense of touch is the name given to a network of nerve endings that reach just about every part of our body. These sensory nerve endings are located just below the skin and register light and heavy pressure on the skin and also differences in temperature. These nerve endings gather information and send it to the brain

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Human and robot sensors quiz

- 5 main senses: vision, hearing, smell, touch and taste
- Human sensor: eyes, ears, nose, skin and tongue
- Additional sensors: temperature sensors, body position sensors, balance sensors and blood acidity sensors

- How they work:
  - Eyes: surrounding light -> relay it to nerve cells -> send signal to brain
  - Ears: sound wave -> air vibrations -> inner ear -> hair cells -> send signal to brain
  - Nose: particle are inhaled into the nose -> nerve cell contact the particles -> send signal to brain
  - Skin: sensor all over the skin are activate and send signals to brain through the nervous system
  - Tongue: particles in food -> hair cell -> send signals to brain through the nervous system

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- Your sensory organs (eyes, ears, nose, tongue, and skin) provide information to your brain so that it can make decisions.
- Several additional sensors in the body that donot notice directly
  - Sensors in the inner ear give the brain information about balance
  - Sensor in muscles inform the brain of our body positions
  - Sensor throughout the body that sense temperature
  - … and more

Question

- Name some sensors used in robot

SCAN ME

https://app.sli.do/event/uJZW4iwCMqaVmud2y6rvJG/embed/polls/69df39d5-37f4-470b-beac-cc2133333c9a

Contents

- Introduction to Robotics
- Robot Sensors
- Desirable features
- Need for Sensors
- Classification of Sensors

What is a robot

- A robot is a machine designed to execute one or more tasks repeatedly, with speed and precision. There are as many different types of robots as there are tasks for them to perform. It is a mechanical or virtual artificial agent, usually an electro-mechanical machine that is guided by a computer program or electronic circuitry.
- A robot can be controlled by a human operator, sometimes from a great distance. But most robots are controlled by computer, and fall into either of two categories: autonomous robots and insect robots. An autonomous robot acts as a stand-alone system, complete with its own computer
- For many people it is a machine that imitates a human—like the androids in Star Wars, Terminator and Star Trek: The Next Generation. However much these robots capture our imagination, such robots still only inhabit Science Fiction.

Classification by JIR

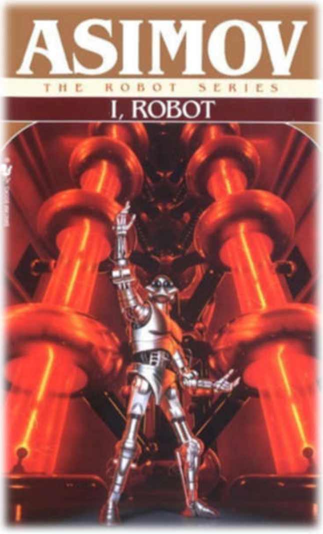
- Japanese Industrial Robot Association (JIRA) :

"A device with degrees of freedom that canbe controlled.":

  - Class 1: Manual handling device
  - Class 2 : Fixed sequence robot
  - Class 3 : Variable sequence robot
  - Class 4 : Playback robot
  - Class 5 : Numerical control robot
  - Class 6 : Intelligent robot

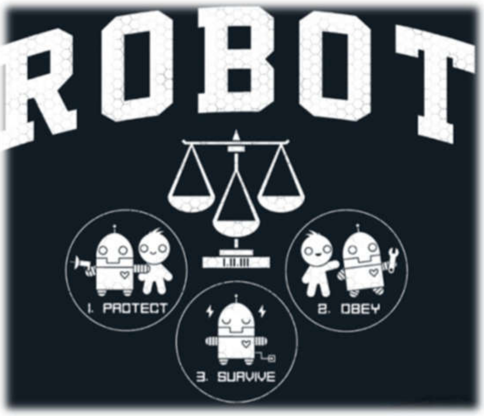
Isaac Asimov

- Was born on January 2, 1920 in America
- An American author and professor of biochemistry at Boston University
- Wrote about 500 popular science books, and most importantly he wrote about robotics and created the Laws of robotics.
- Asimov was very famous for his writing and ideas on robotics. He's first short story to be sold is "Marooned Off Vesta" And Asimov would later be credited with coming up with the term "robotics"



The Three Laws of Robotics

Created by the science fiction author Isaac Asimov (introduced in his 1942 novel "I, Robot")



- A robot may **not injure a human being** or, through inaction, allow a human being to come to harm.
- A robot must **obey the orders** given to it by human beings except where such orders would conflict with the First Law.
- A robot must **protect its own existence** as long as such protection does not conflict with the First or Second Laws.

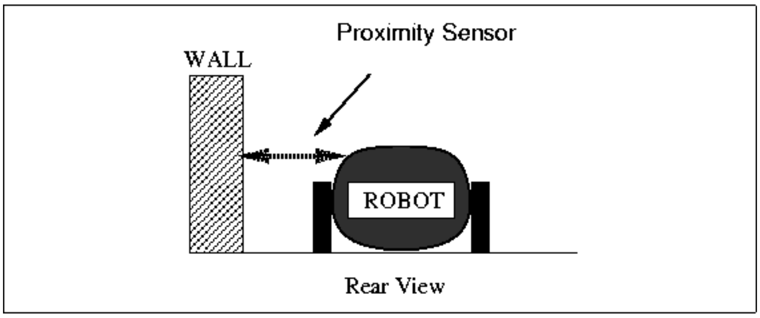
3 Robotic Laws	Benefits	Disadvantages
1) A robot may not injure a human being or, through inaction, allow a human being to come to harm	This law protects human and insure safety of human.	Robots may not work perfectly and they may not obey human's orders.
2) A robot must obey the orders given it by human beings, except where such orders would conflict with the First Law.	It is better for human to control robots to prevent bad consequence.	It seems like a discrimination against the robots. Human have more rights than the robots, so it is not fair for robots.
3) A robot must protect its own existence as long as such protection does not conflict with the First or Second Laws.	It gives robots some rights to protect themselves.	If robots do not obey human's order and break the three law of robotics. There are no penalties to the robots or they cannot blame lame the person who created that robot.

*While these "rules" are commonly used by science fiction authors and screenwriters, they are **NOT** actual laws that must be followed by engineers in their robotic inventions.*

Robotic Sensing

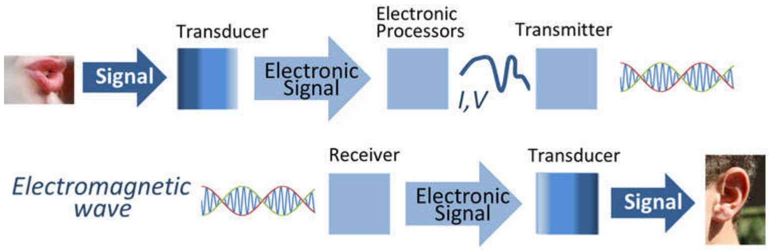
- A branch of robotics science intended to give robots sensing capabilities, so that robots are more human-like.
- Robotic sensing mainly gives robots the ability to see, touch, hear and move and uses algorithms that require environmental feedback.
- The use of sensors in robots has taken them into the next level of creativity. Most importantly, the sensor shave increased the performance of robots to a large extent. It also allows the robots to perform several functions like a human being.

- The robot uses sensors to interact with its environment.
- There are a variety of sensors used for a variety of purposes: Pressure, rotation, temperature, smoke, tilt, vibration, light, proximity and so on.



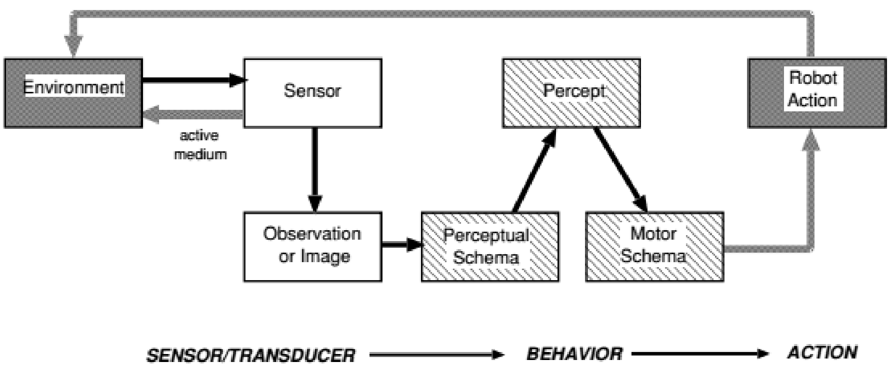
Overview

- What are Sensors?
- Detectable Phenomenon
- Physical Principles – How Do Sensors Work?
- Need for Sensors
- Choosing a Sensor
- Sensor Descriptions



Sensors in Robot

- A model of sensing



- Sensing
- Sensor

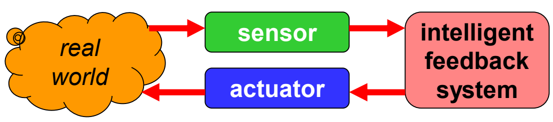
Need for Sensors

- Sensors are omnipresent. They embedded in our bodies, automobiles, airplanes, cellular telephones, radios, chemical plants, industrial plants and countless other applications.

- Without the use of sensors, there would be no automation !!
  - Imagine having to manually fill Poland Spring bottles

Transducers

- Transducer
  - a device that converts a primary form of energy into a corresponding signal with a different energy form
    - Primary Energy Forms: mechanical, thermal, electromagnetic, optical, chemical, etc.
  - take form of a **sensor** or an **actuator**
- Sensor (e.g., thermometer)
  - a device that detects/measures a signal or stimulus
  - acquires information from the "real world"
- Actuator (e.g., heater)
  - a device that generates a signal or stimulus

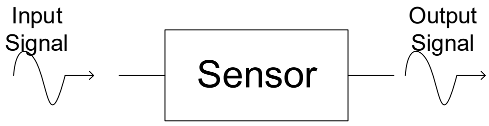


- **Sensors** are devices that responds to a physicalstimulus he at, light, sound, pressure, magnetism, motion, etc. and convert that into an electrical signal. They perform an input function.
- **Actuators** are devices which perform an output function and are used to control some external device, for example movement. Coverts electrical energy tomecha nical movement (motors)
- Both sensors and actuators are collectively known as **Transducers**
- Transducers are devices used to convert energy of onekind into energy of another kind

- **Robotic sensors** are used to estimate a robot's condition and environment. These signals are passedto a controller to enable appropriate behavior.
- They sense and measure geometric and physical properties of robots and the surroundingenvironment
  - Position, orientation, velocity, acceleration
  - Distance, size
  - Force, moment
  - Temperature, Luminance,
- Sensors in robots are based on the functions of human sensory organs.
- Robots require extensive information about their environment in order to function effectively

What are Sensors?

- American National Standards Institute (ANSI) Definition
  - A device which provides a usable output in response to a specified measurand



- A sensor acquires a physical parameter and converts it into a signal suitable for processing (e.g. optical, electrical, mechanical)

- Example:

https://www.elprocus.com/difference-between-sensor-and-transducer/

Typical Sensors

Features of Sensors

- **Accuracy:** Accuracy should be high. How close outputto the true value is the accuracy of the device.
- **Precision :** There should not be any variations in the sensed output over a period of time precision of thesensor should be high.
- **Operating Range:** Sensor should have wide range of operation and should be accurate and precise overthis entire range.
- **Speed of Response:** Should be capable of responding to the changes in the sensed variable in minimumtime.

- **Calibration:** Sensor should be easy to calibrate, timeand trouble required to calibrate should beminimum. I t should not require frequent recalibration.
- **Reliability:** It should have high reliability. Frequentfailure should not happen.
- **Cost and Ease of operation:** Cost should be as lowas possible, installation, operation and maintenanceshould be easy and should not required skilled or highly trained persons.

Choosing a Sensor

Environmental Factors	Economic Factors	Sensor Characteristics
Temperature range	Cost	Sensitivity
Humidity effects	Availability	Range
Corrosion	Lifetime	Stability
Size		Repeatability
Overrange protection		Linearity
Susceptibility to EM interferences		Error
Ruggedness		Response time
Power consumption		Frequency response
Self-test capability		

Classification of Sensors

- Proprioceptive (Internal state) v.s. Exteroceptive
  - Measure values internally to the system (battery level, wheel position, joint angle…)
  - Observation of environments, objects
- Active vs Passive
  - Emitting energy into the environment, e.g., infrared, radar, sonar
  - Passively receive energy to make observation: camera
- Contact vs. non-contact
- …

Common Sensors and Transducers

Stimulus	Quantity
Acoustic	Wave (amplitude, phase, polarization), Spectrum, Wave Velocity
Biological & Chemical	Fluid Concentrations (Gas or Liquid)
Electric	Charge, Voltage, Current, Electric Field (amplitude, phase, polarization), Conductivity, Permittivity
Magnetic	Magnetic Field (amplitude, phase, polarization), Flux, Permeability
Optical	Refractive Index, Reflectivity, Absorption
Thermal	Temperature, Flux, Specific Heat, Thermal Conductivity
Mechanical	Position, Velocity, Acceleration, Force, Strain, Stress, Pressure, Torque

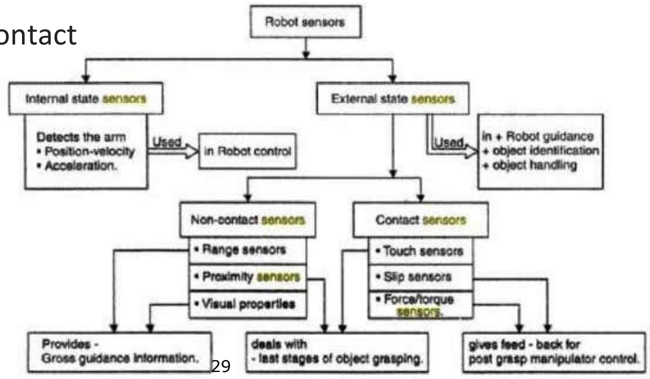
Quantity being Measured	Input Device (Sensor)	Output Device (Actuator)
Light Level	Light Dependant Resistor (LDR) Photodiode Photo-transistor Solar Cell	Lights & Lamps LED's & Displays Fibre Optics
Temperature	Thermocouple Thermistor Thermostat Resistive Temperature Detectors	Heater Fan
Force/Pressure	Strain Gauge Pressure Switch Load Cells	Lifts & Jacks Electromagnet Vibration
Position	Potentiometer Encoders Reflective/Slotted Opto-switch LVDT	Motor Solenoid Panel Meters
Speed	Tacho-generator Reflective/Slotted Opto-coupler Doppler Effect Sensors	AC and DC Motors Stepper Motor Brake
Sound	Carbon Microphone Piezo-electric Crystal	Bell Buzzer Loudspeaker

Need for Sensors

- Sensors are needed in robotics in order to makethem automated.
- Without sensors, a robot is, in essence, blind and deaf.
- Sensors allow a robot to collect information from the surrounding environment, in order to interact with it.
- Humanoid robots need a multitude of these sensors in order to mimic the capabilities of their living counterparts (human).
- Sensors are important in creating robots which areefficient in their appointed purpose.
- The inclusion of sensors is imperative to theirautomation. However, developers need to take carewhen choosing which sensors to incorporate in the design.
- Sensor type, sensitivity, accuracy, and position are all important factors for the success of the robot.

Classification of Sensors

- There are a wide variety of sensors used in robots
- Sensors can be classified using two important functional axes
- Proprioceptive / exteroceptive (internal/external)
- passive/active
- Contact/noncontact



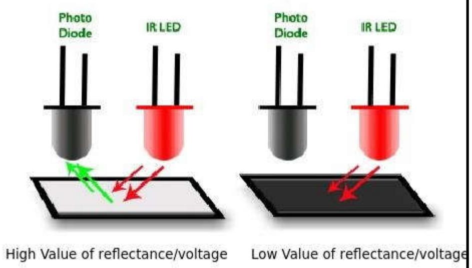
- **Proprioceptive** sensors measure valuesinternal to the robot; for example, motor speed, wheel load, robot arm joint angles, and battery voltage.
- **Passive** sensors measure ambientenvironment energy ente ring the sensor.Examples of passive sensors include
  - temperature probes,
  - microphones,
  - and CCD or CMOS cameras.

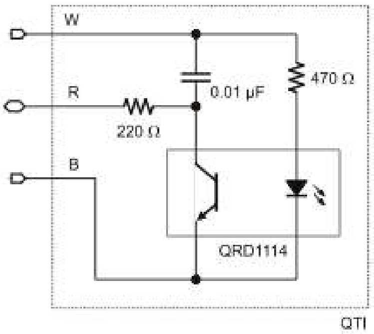
- Industrial manufacturing robots need sensors to allow them to operate efficiently for pick and place of objects without crushing or dropping them -> torque sensors, which monitor and control rotational forces
- Acoustical and piezoelectric sensors allow robots to identify prominent sounds, such as commands, in an area with background noise.
- Robots can incorporate pre-programmed outputs based upon the commands heard. This could be especially useful for field work, or robots in noisy environments.
- The simplest sensors are switch sensors. There are three types of switch sensor; contact, limitand shaft encoder sensors. They work without processing.

- Light sensors can also be used in multiple ways.They give a robot the ability to see.
- Light sensors allow robots to measure light intensity, differential intensity, and break-beam, i.e. the sudden reduction of intensity.
- These light sensors can be applied in different positions and directions depending on therobot’s intended purpose.

Ex. Line Sensor

- QRD1114: infrared (IR) reflective sensor to determine the reflectivity of the surface below it.
- When over the black playing field, the reflectivity is very low
- When over the white border, the reflectivity is very high and will cause a different reading from the sensor.





Khoa Điện tử - Viễn thông
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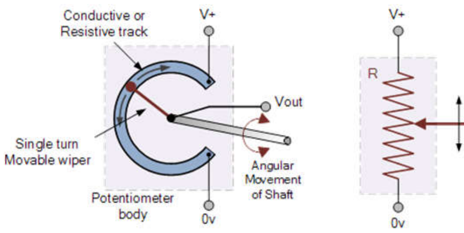
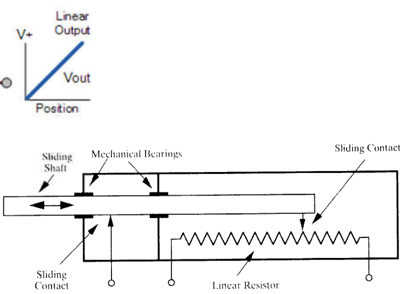
Ex. Position sensors

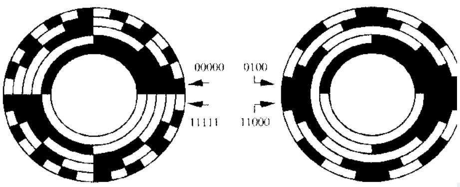
- Rotary



Linear

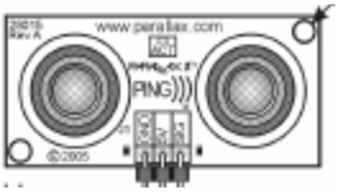


- Optical shaft encoder



Ex. Ultrasonic Proximity Sensor

PING Ultrasonic Range Finder

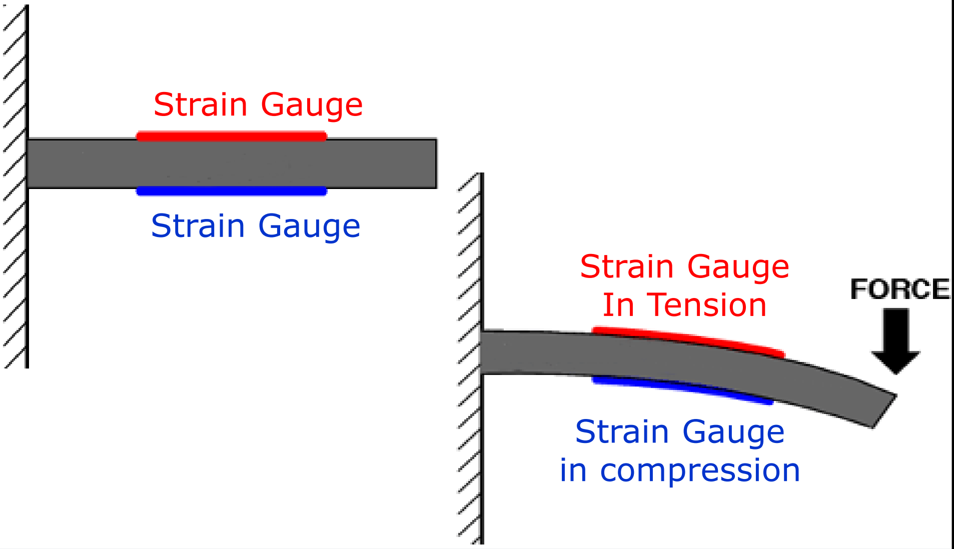


•PING ultrasonic distance sensor provides precise distance measurements from about 2 cm (0.8 inches) to 3 meters (3.3 yards).

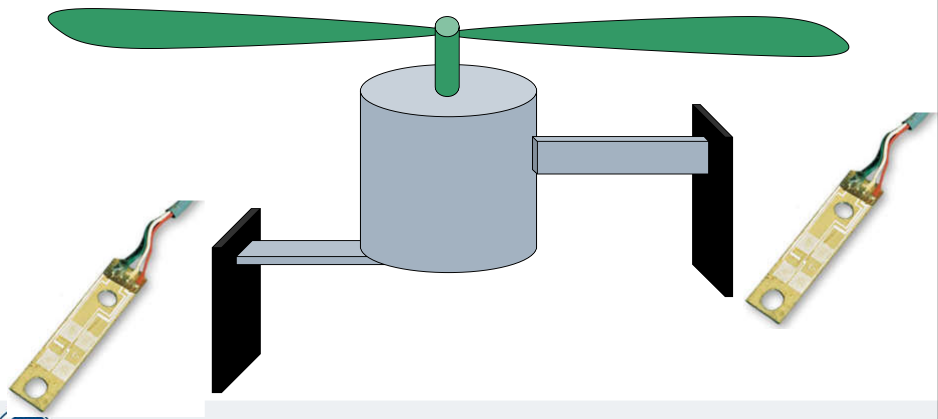
•It works by transmitting an ultrasonic burst and providing an output pulse that corresponds to the time required for the burst echo to return to the sensor.

•By measuring the echo pulse width the distance to target can easily be calculated.

Ex. Bending Beam Load Cell



Measuring Thrust & Torque



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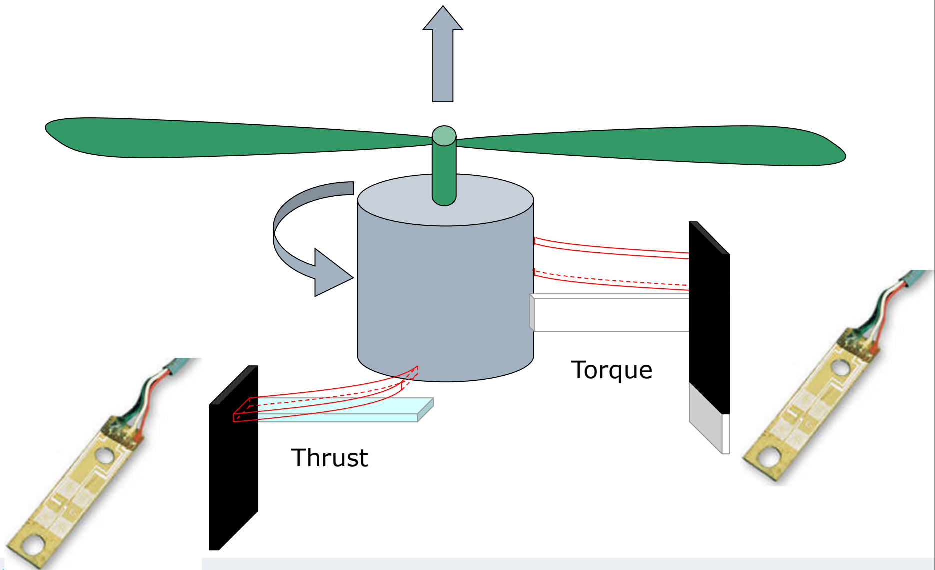
Measuring Thrust & Torque



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Course Objectives

- Kiến thức
  - Giới thiệu tổng quan về kỹ thuật đo lường và cảm biến
  - Các nguyên lý, kỹ thuật chuyển đổi tín hiệu từ môi trường thành tín hiệu điện
  - Thiết kế, cấu tạo, hoạt động và ứng dụng của một số cảm biến
- Kỹ năng
  - Phân tích thiết kế, triển khai hệ thống cảm biến điện tử
  - Các công cụ tính toán và phân tích như Matlab, Ansys, Comsol, …

Summary

- Introduction to Robotics
- Robot Sensors
- Desirable features
- Need for Sensors
- Classification of Sensors

Textbook and references

- Textbook: Introduction to sensors for electrical and mechanical engineers, Novák, Martin
- References
  - Sensors and Actuators A: Physical - Journals | Elsevier https://www.journals.elsevier.com/sensors-and-actuators-a-physical
  - Sensors and Actuators B: Chemical - Journals | Elsevier https://www.journals.elsevier.com/sensors-and-actuators-b-chemical
  - IEEE Sensors Journal https://ieeexplore.ieee.org/xpl/RecentIssue.jsp?punumber=7361
  - …

(https://sci-hub.shop/)

Software and Tool

- Matlab
- Comsol

Home works and Essays

- Home works: Website Courses.vnu.edu.vn
- Essay: 10 group of about 5 students